# The Fourier Series: Continuous-Time Periodic Signals



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# Key Concepts

1) The Fourier series applies to continuous-time periodic signals.

2) The Fourier series is used to analyze periodic signals and their interaction with LTI systems.

3) The Fourier series expansion

$$x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{jk\Omega_o t}$$

where $x(t)$ has fundamental period $T$ and fundamental frequency $\Omega_o = 2\pi/T$. $X[k]$ are the Fourier series coefficients.

4) Fourier series coefficients

$$X[k] = \frac{1}{T}\int_{<T>} x(t)e^{-jk\Omega_o t}dt$$

where $< T >$ denotes integration over any interval of length $T$.

5) The Fourier series expresses $x(t)$ as a weighted sum of sinusoids having harmonically related frequencies.

6) The infinite series for $x(t)$ does not converge for all signals.

a) If $x(t)$ is square integrable ($\int_{<T>} |x(t)|^2 dt < \infty$, then the mean-squared error between $x(t)$ and the infinite sum is zero.

b) If $x(t)$ satisfies the Dirichlet conditions, then the infinite sum converges point wise to $x(t)$.

   i) $x(t)$ is bounded.

   ii) $x(t)$ has a finite number of minima and maxima in one period.

   iii) $x(t)$ has a finite number of discontinuities in one period.

c) If $x(t)$ satisfies the Dirichlet conditions but is not continuous, the sum converges to the midpoint of the left and right limits of $x(t)$ at the discontinuities.

7) Trigonometric Fourier series for real-valued $x(t)$

a) If $x(t)$ is real-valued, then the Fourier series coefficients are conjugate symmetric $X[-k] = X^*[k]$.

b) The Fourier series may be rewritten

$$x(t) = X[0] + \sum_{k=1}^{\infty} B[k]\cos(k\Omega_o t) + A[k]\sin(k\Omega_o t)$$

where $B[k] = X[k] + X[-k]$ and $A[k] = j(X[k] - X[-k])$ or

$$B[k] = \frac{2}{T} \int_{<T>} x(t) \cos(k\Omega_o t) dt$$

$$A[k] = \frac{2}{T} \int_{<T>} x(t) \sin(k\Omega_o t) dt$$

# Video Screenshots

# Fourier Series - representing continuous-time periodic signals using sinusoids



- Sinusoids have simple interaction with LTI systems
- Analysis tool. signal-system interaction, waveform analysis

Continuous-time signal → use continuous-time sinusoids

Period $T$ → use sinusoids with period $T$ $\quad \Omega_k = k\frac{2\pi}{T}, \quad \Omega_0 = \frac{2\pi}{T}$

$$x(t) = \sum_{k=-\infty}^{\infty} X[k]\, e^{jk\Omega_0 t}$$

$e^{jk\Omega_0 t}$: sinusoid with frequency $k\Omega_0$

$X[k]$: weight

$$X[k] = \frac{1}{T}\int_{\langle T\rangle} x(t)\, e^{-jk\Omega_0 t}\, dt$$

$\langle T\rangle$: any length $T$ interval $[0, T)$, $[-\frac{T}{2}, \frac{T}{2})$

Verify by substituting $\sum_{k=-\infty}^{\infty} X[k]e^{jk\Omega_0 t}$ for $x(t)$ in integrand

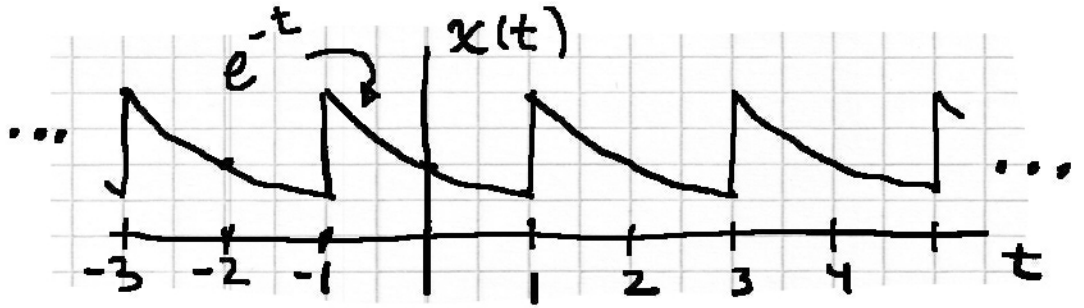
Notation: $x(t) \xleftrightarrow{FS;\Omega_0} X[k]$

Convergence of infinite sum:

If $\int_0^T |x(t)|^2 dt < \infty \rightarrow \frac{1}{T}\int_0^T |x(t) - \sum_{k=-\infty}^{\infty} X[k] e^{jk\Omega_0 t}|^2 dt = 0$ Zero mean-square error

Dirichlet conditions guarantee pointwise convergence
- at discontinuities sum converges to midpoint

Example:



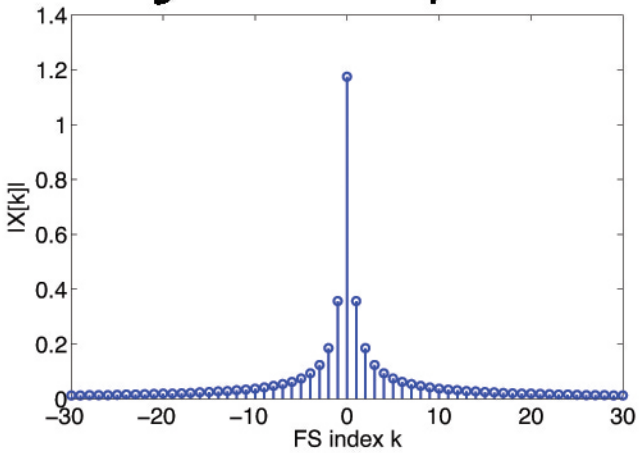

$T = 2$ sec

$\Omega_0 = \frac{2\pi}{T} = \pi$ $\frac{\text{rads}}{\text{sec}}$

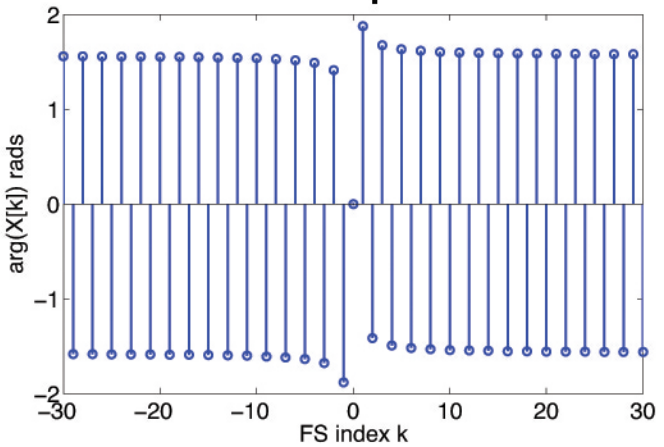
$$X[k] = \frac{1}{2}\int_{-1}^{1} e^{-t} e^{-jk\pi t} dt = \frac{1}{2}\int_{-1}^{1} e^{-(1+jk\pi)t} dt = \frac{-1}{2(1+jk\pi)} e^{-(1+jk\pi)t}\Big|_{-1}^{1}$$

$$= \frac{e^{(1+jk\pi)} - e^{-(1+jk\pi)}}{2 + j2k\pi}$$

## Magnitude Spectrum




## Phase Spectrum




## Example: Find $x(t)$

$$X[k] = \begin{cases} 1, & k = \pm 2 \\ -j, & k = 4 \\ j, & k = -4 \\ 2+4j, & k = 8 \\ 2-4j, & k = -8 \\ 0, & \text{otherwise} \end{cases} \qquad T = 3 \text{ sec}$$

1) $\Omega_0 = \frac{2\pi}{3}$ rads/sec

$$x(t) = e^{j2\Omega_0 t} + e^{-j2\Omega_0 t} - j e^{j4\Omega_0 t} + j e^{-j4\Omega_0 t} + (2+4j) e^{j8\Omega_0 t} + (2-4j) e^{-j8\Omega_0 t}$$

$$= 2\cos(2\Omega_0 t) + 2\sin(4\Omega_0 t) + 2\left(e^{j8\Omega_0 t} + e^{-j8\Omega_0 t}\right) + 4j\left(e^{j8\Omega_0 t} - e^{-j8\Omega_0 t}\right)$$

$$= 2\cos(2\Omega_0 t) + 2\sin(4\Omega_0 t) + 4\cos(8\Omega_0 t) - 8\sin(8\Omega_0 t)$$

# Trigonometric Fourier Series for real $x(t)$



express $x(t)$ using sines/cosines

Real $x(t) \Leftrightarrow X[-k] = X^*[k]$ (conjugate symmetry)

$$X^*[k] = \frac{1}{T}\int_{<T>} \left[x(t)e^{-jk\Omega_0 t}\right]^* dt = \frac{1}{T}\int_{<T>} x(t)e^{-j(-k)\Omega_0 t}dt = X[-k]$$

$$x(t) = \sum_{k=-\infty}^{\infty} X[k]e^{jk\Omega_0 t} \rightarrow \text{Define } X[k] = \frac{B[k]}{2} - j\frac{A[k]}{2};\ X[-k] = \frac{B[k]}{2} + j\frac{A[k]}{2}$$

$$x(t) = X[0] + \sum_{k=1}^{\infty} X[k]e^{jk\Omega_0 t} + X[-k]e^{-jk\Omega_0 t}$$

$$= X[0] + \sum_{k=1}^{\infty} B[k]\left(\frac{e^{jk\Omega_0 t} + e^{-jk\Omega_0 t}}{2}\right) + A[k]\left(\frac{e^{jk\Omega_0 t} - e^{-jk\Omega_0 t}}{2j}\right)$$

$$x(t) = X[0] + \sum_{k=1}^{\infty} B[k]\cos(k\Omega_0 t) + A[k]\sin(k\Omega_0 t);$$

$$X[0] = \frac{1}{T}\int_{<T>} x(t)\,dt$$

$$B[k] = \frac{2}{T}\int_{<T>} x(t)\cos(k\Omega_0 t)dt$$

$$A[k] = \frac{2}{T}\int_{<T>} x(t)\sin(k\Omega_0 t)dt$$

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